

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

Comparative Analysis of Workplace and Urban Charging Ecosystems

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Data Landscape & Preprocessing

Dataset	Type	Granularity	Key Variable
ACN	Workplace charging	Session-level	Sessions
UrbanEV	Urban occupancy	5-min intervals	Occupancy

ACN Pipeline

- Timestamp parsing
- Duration computation
- Idle-time engineering
- Hourly aggregation

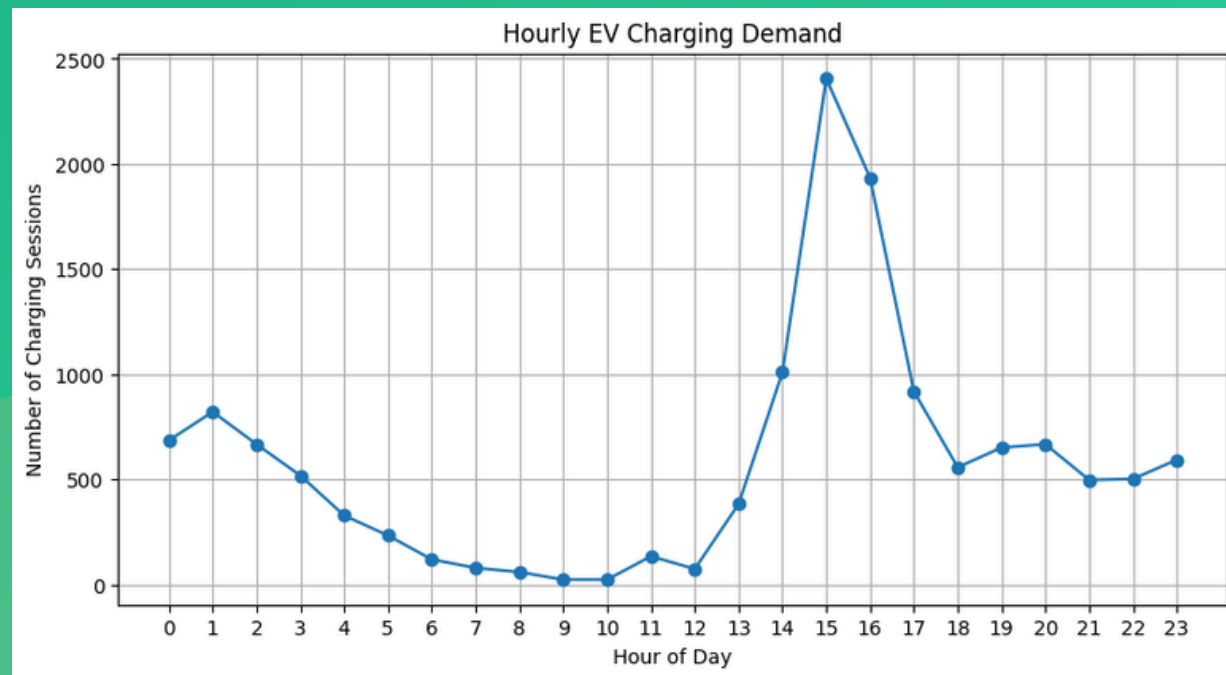
UrbanEV Pipeline

- Timestamp reconstruction
- Matrix-to-long conversion
- Occupancy alignment
- Temporal feature engineering

Step	Description
Raw Data	ACN & UrbanEV datasets
Cleaning	Timestamp correction, missing value handling
Feature Engineering	Lag variables, utilization metrics
Aggregation	Hourly occupancy & demand creation
Forecasting	Random Forest demand prediction
Optimization	Dynamic congestion-aware pricing
Monitoring	Pricing efficiency & congestion tracking

Demand Behavior & Congestion Insights

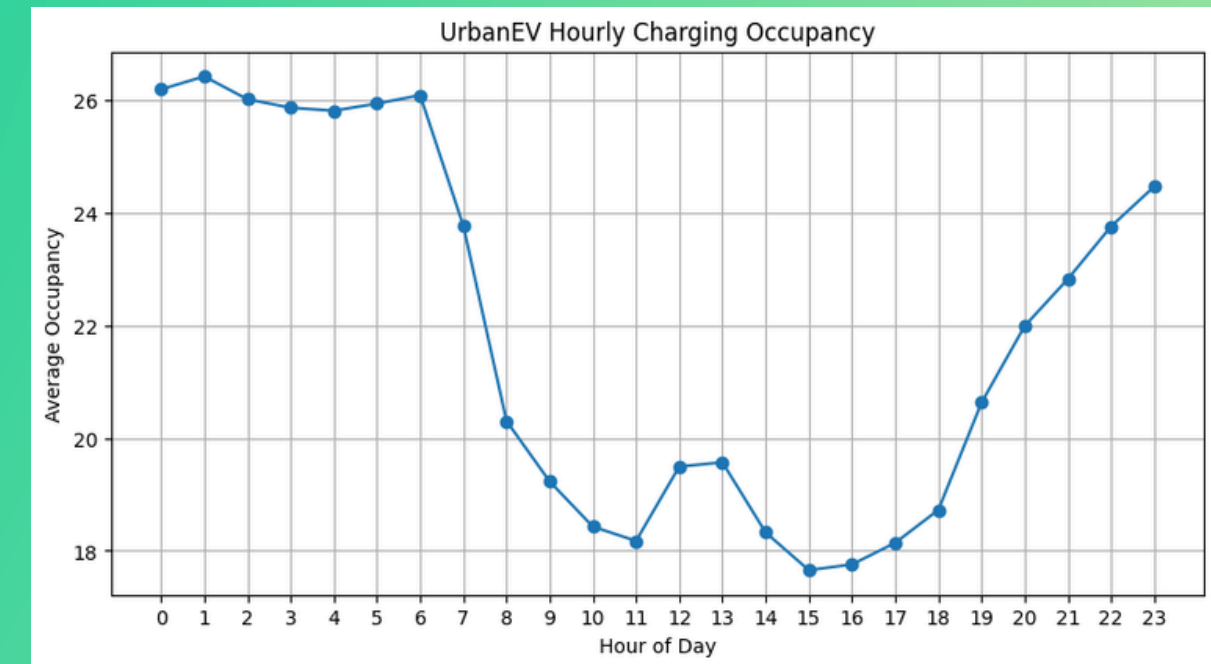
ACN



ACN Key Findings

- Demand peaks at: 15:00–16:00
- Strong weekday congestion
- Long idle occupation periods
- Highly concentrated workplace charging behavior

UrbanEV



UrbanEV Key Findings

- Peak occupancy occurs overnight
- Weekend occupancy remains elevated
- Persistent city-scale charging pressure
- Smoother utilization dynamics

Demand Prediction Modeling & Results

Random Forest Regressor

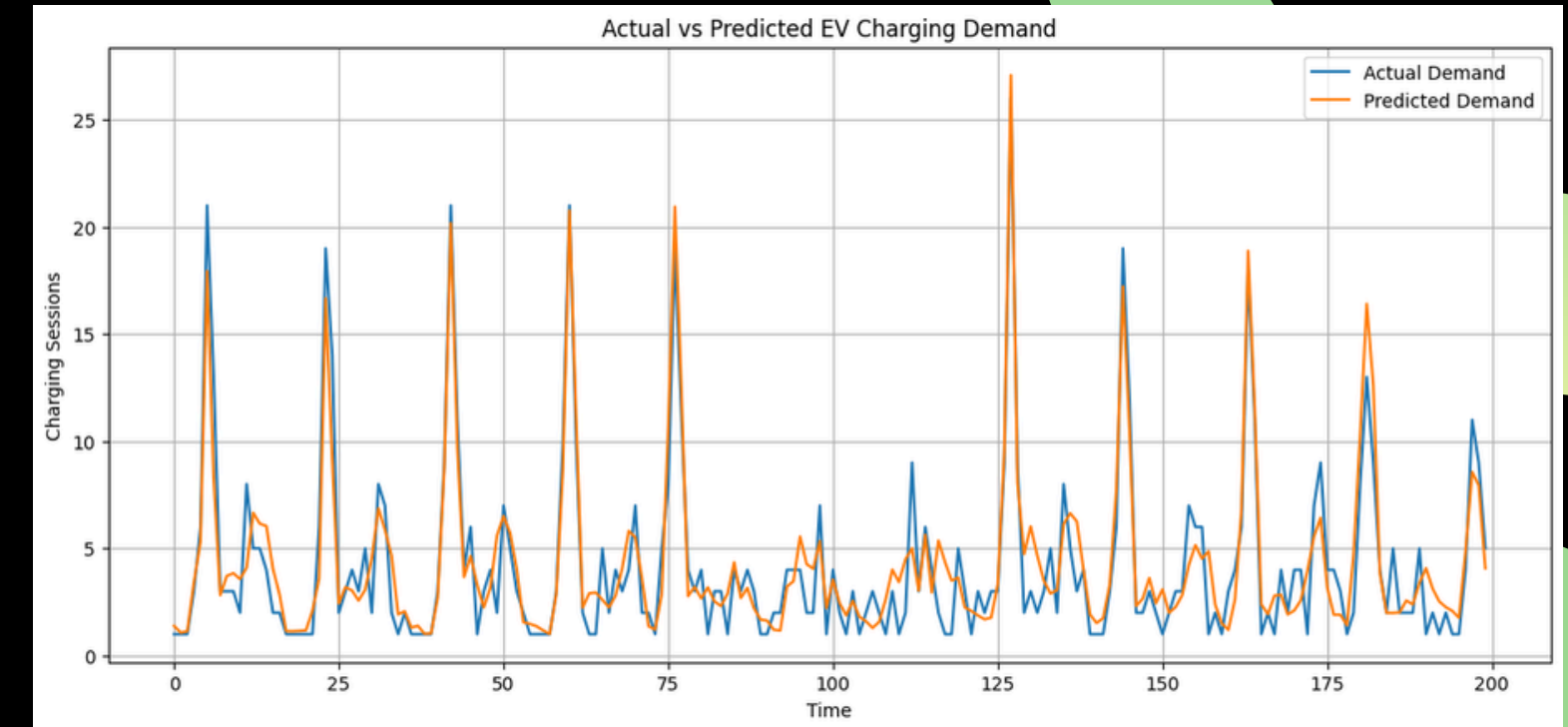
Features

- Lag variables
- Rolling averages
- Hour/day/month indicators

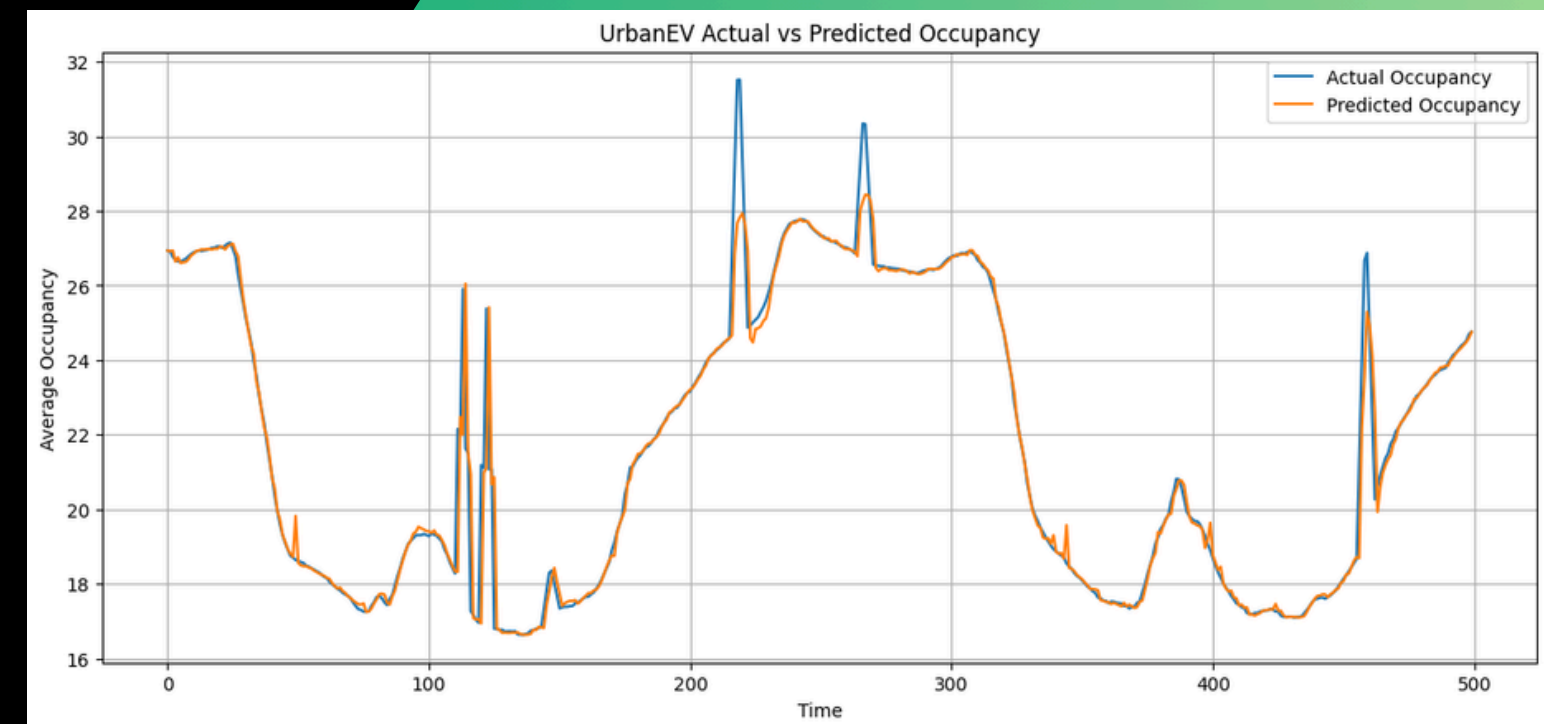
Dataset	MAE	RMSE	R ²
ACN	1.37	1.99	0.67
UrbanEV	0.14	0.44	0.985

Urban occupancy dynamics are substantially more predictable than individual charging session arrivals.

Actual vs Predicted ACN



Actual vs Predicted UrbanEV



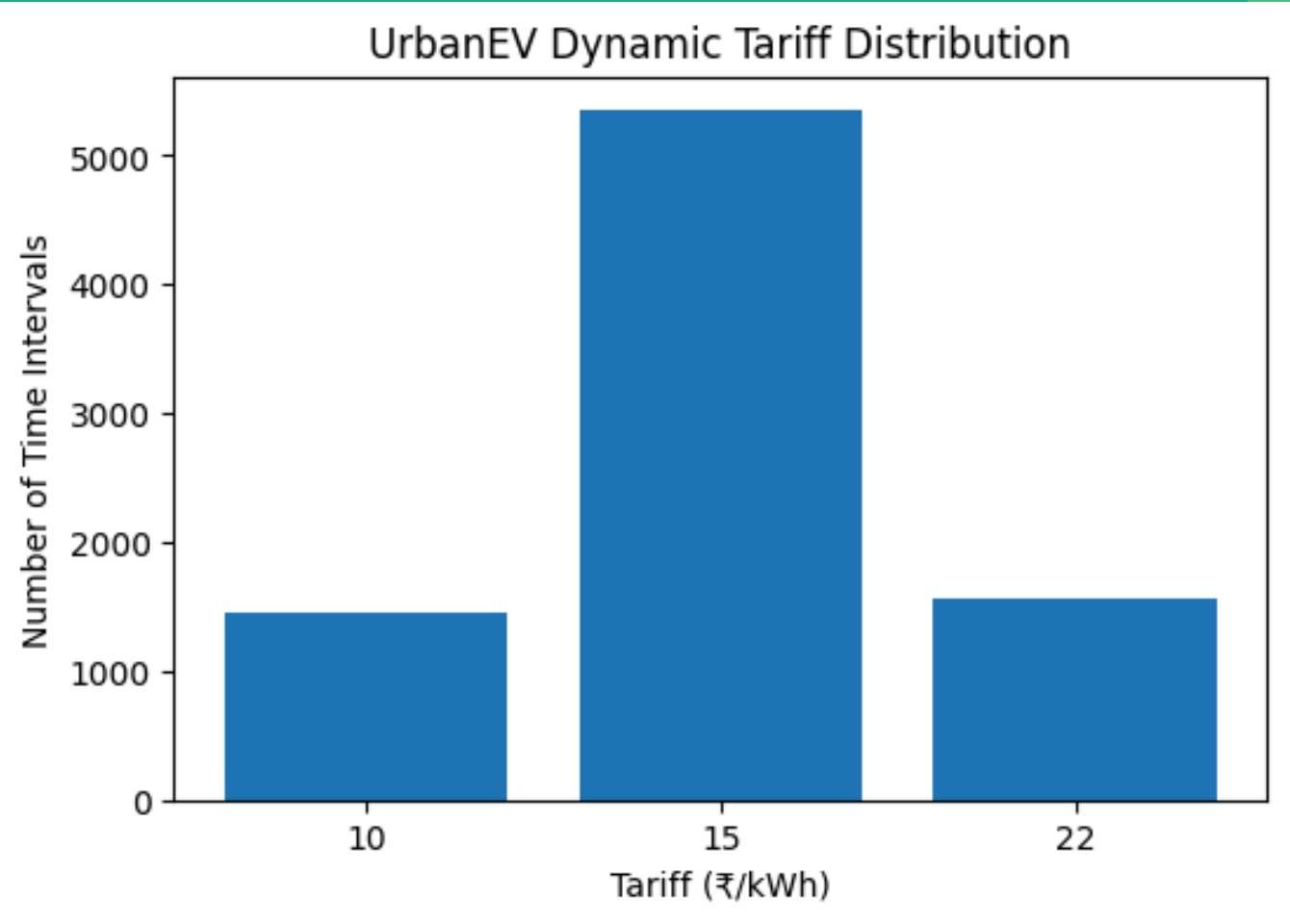
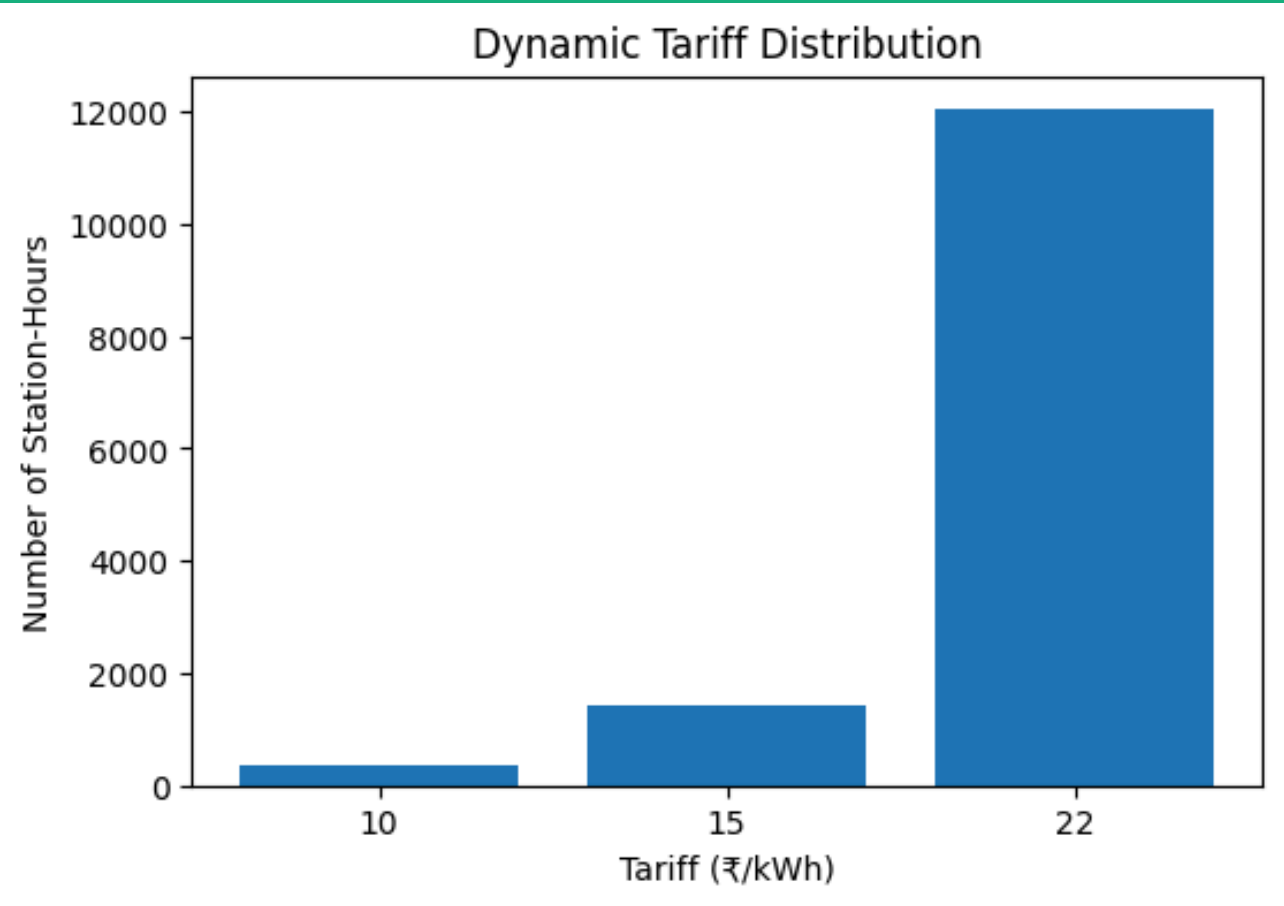
Dynamic Tariff Optimization

Pricing Table

Congestion Level	Tariff
High Congestion	₹22/kWh
Normal	₹15/kWh
Low Utilization	₹10/kWh

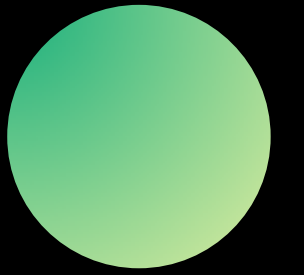
Revenue Outcomes

Dataset	Baseline	Dynamic	Uplift
ACN	₹1.85M	₹2.68M	45%
UrbanEV	₹2.74M	₹2.91M	6%



**DYNAMIC PRICING
EFFECTIVENESS DEPENDS
STRONGLY ON DEMAND
CONCENTRATION AND
CONGESTION INTENSITY.**

Monitoring Agent & Operational Feedback Loop



Monitoring Components

- **CONGESTION CLASSIFICATION ACROSS CHARGING INTERVALS**
- **IDENTIFICATION OF OVERLOADED AND UNDERUTILIZED STATIONS**
- **PRICING EFFICIENCY TRACKING UNDER ADAPTIVE TARIFFS**
- **MONITORING OF SURGE, STANDARD, AND DISCOUNT TARIFF ACTIVATION**

ACN NETWORK

- **PERSISTENT HIGH CONGESTION OBSERVED**
- **SURGE PRICING ACTIVATED FREQUENTLY**
- **WORKPLACE CHARGING CREATED CONCENTRATED DEMAND SPIKES**

URBANEV NETWORK

- **BALANCED TARIFF ACTIVATION DISTRIBUTION**
- **STABLE UTILIZATION DYNAMICS ACROSS TIME**
- **MODERATE CONGESTION WITH SMOOTHER OCCUPANCY BEHAVIOR**

CONTINUOUS MONITORING ENABLES ADAPTIVE PRICING SYSTEMS TO DYNAMICALLY RESPOND TO CHANGING CHARGING DEMAND CONDITIONS AND IMPROVE INFRASTRUCTURE UTILIZATION EFFICIENCY.



Business, Operational & Policy Implications

Implications

COMPARISON

Metric	ACN	UrbanEV
Charging Type	Workplace	Urban Residential
Peak Demand	Afternoon	Overnight
Forecastability	Moderate	Very High
Revenue Uplift	45%	6%
Pricing Volatility	High	Moderate

BUSINESS

- HIGHER MONETIZATION EFFICIENCY
- INFRASTRUCTURE OPTIMIZATION
- BETTER REVENUE CAPTURE

OPERATIONAL

- CONGESTION MITIGATION
- CHARGER BALANCING
- DEMAND REDISTRIBUTION

POLICY

- DYNAMIC TARIFF REGULATION
- GRID-AWARE CHARGING INCENTIVES
- URBAN CHARGING PLANNING

CONCLUSION

DYNAMIC TARIFF OPTIMIZATION EFFECTIVENESS DEPENDS STRONGLY ON CHARGING ECOSYSTEM STRUCTURE. WORKPLACE CHARGING NETWORKS BENEFIT SUBSTANTIALY FROM CONGESTION-AWARE SURGE PRICING, WHILE URBAN CHARGING ECOSYSTEMS EXHIBIT MORE STABLE OCCUPANCY DYNAMICS THAT FAVOR MODERATE ADAPTIVE PRICING AND LONG-TERM INFRASTRUCTURE BALANCING STRATEGIES.

THANK YOU